

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA1017	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	LOSHINI.L	Reg. No.:	192571015
Date:	22/8/26	Duration:	60 minutes

Code of Conduct

I LOSHINI.L 192571015__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: LOSHINI.L_____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

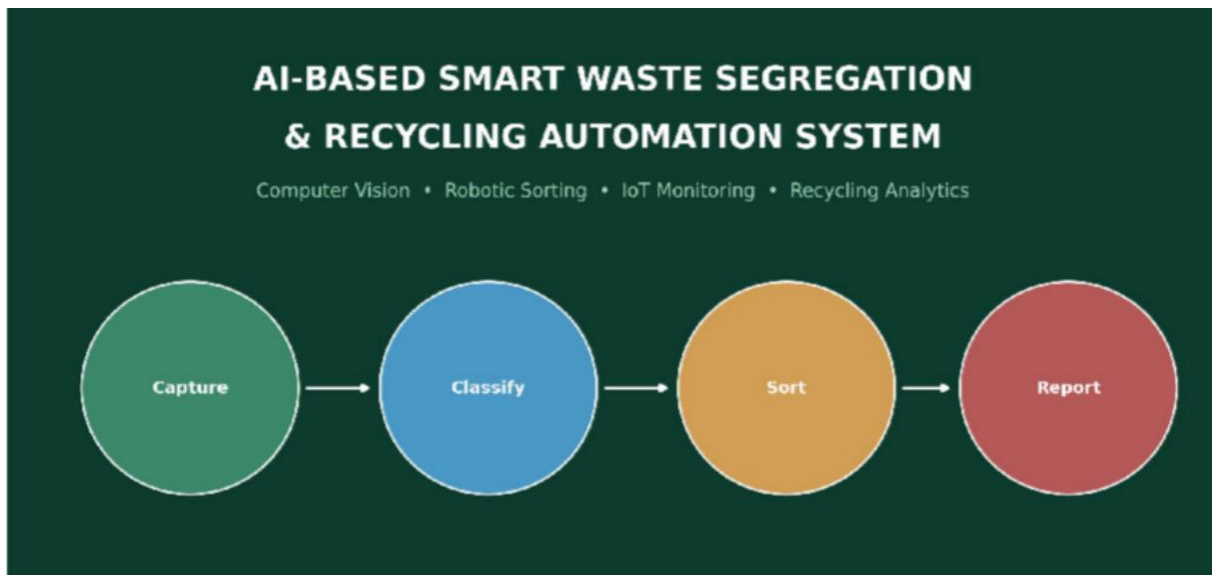
Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5

Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25



AI-Based Smart Waste Segregation and Recycling Automation System

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Abstract

Improper segregation of municipal solid waste remains a major bottleneck in recycling efficiency worldwide, with manual sorting processes producing inconsistent results and high operational costs. This literature survey reviews recent research (2024–2026) on AI-powered waste segregation and recycling automation, covering convolutional neural network (CNN) classifiers, YOLO-based object detectors, robotic sorting arms, IoT-enabled smart bins, and analytics dashboards. Six representative studies are reviewed in detail and compared on methodology, dataset/tools, reported accuracy, and limitations. The survey identifies recurring research gaps — including limited dataset diversity, poor real-world/cluttered-scene generalisation, high robotic deployment cost, and weak integration between classification, sorting and reporting layers — and proposes a unified system architecture and future research directions that directly address the assigned problem statement of building an AI-powered waste segregation system with computer vision, robotic sorting, recycling-statistics monitoring, and automated report generation.

1. Introduction and Problem Statement

Rapid urbanisation and rising consumption have sharply increased the volume of municipal solid waste generated every year. A large share of potentially recyclable material is lost because household and facility-level segregation is inconsistent, and centralised waste-processing plants still depend heavily on manual sorting. Manual sorting is slow, labour-intensive, exposes workers to health hazards, and produces variable results depending on worker fatigue and training — all of which directly reduce recycling efficiency and increase landfill dependence.

The assigned problem statement calls for an AI-powered waste segregation system that (i) identifies different categories of waste using computer vision, (ii) automates sorting through robotic mechanisms, (iii) monitors recycling statistics, and (iv) generates waste-management reports, with the overall goal of improving recycling efficiency while reducing environmental pollution. To address this problem responsibly, it is first necessary to review how current research has approached each of these four sub-problems, and to identify what remains unresolved.

This report follows the prescribed literature-survey structure: individual review of selected papers, a comparative analysis table, an assessment of limitations and challenges, identification of research gaps, and recommended future directions, followed by a proposed system overview and consistent referencing.

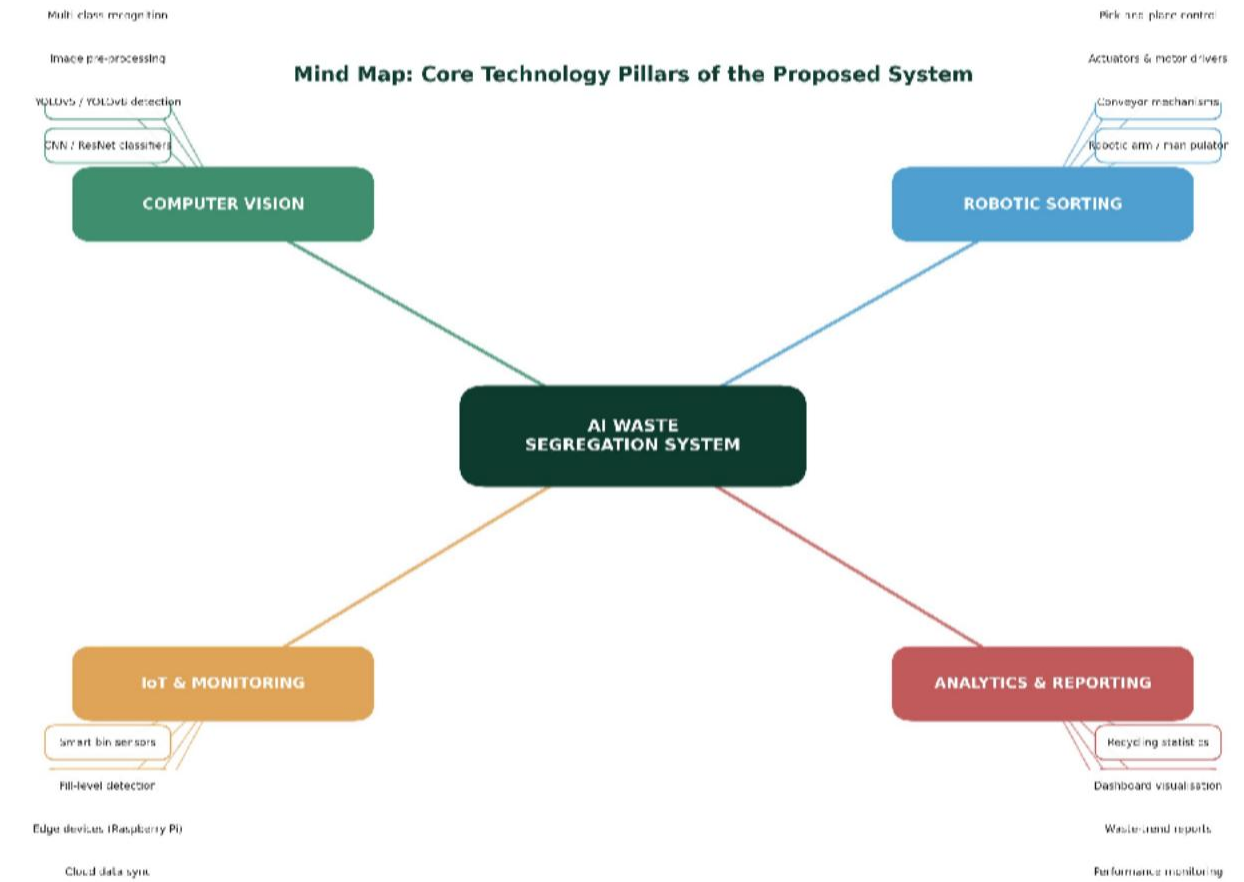


Figure 1. Mind map of the four core technology pillars relevant to the assigned problem statement.

2. Literature Survey

2.1 Intelligent Waste Sorting for Urban Sustainability Using Deep Learning (Scientific Reports, 2025)

This study develops a ResNet-based CNN classifier trained on 15,535 images spanning twelve waste categories, with data augmentation used to address class imbalance. The objective was to build a generalisable image-classification model for organic, recyclable and hazardous waste. Evaluated on precision, recall, F1-score and accuracy, the model reached a classification accuracy of 98.16%, outperforming earlier CNN architectures. The authors note that most prior models were trained on unbalanced or non-diversified datasets, limiting real-world applicability — a gap their augmented, twelve-class dataset partly addresses.

2.2 AI-Based Smart Bin for Efficient and Sustainable Waste Classification (World Journal of Environmental Engineering, 2024)

This paper presents a robotic smart-bin prototype using a VGG16 deep-learning model to classify wet, dry and electronic waste. Hardware consists of a Logitech C920 camera and Raspberry Pi 4 for on-device inference, paired with DC motors and a motor driver for physical sorting. The system achieved up to 98% sorting efficiency in point-of-disposal testing, demonstrating that lightweight, real-time classification combined with simple actuation can be deployed at the individual-bin scale rather than only at centralised facilities.

2.3 Real-Time Solid Waste Sorting Using a Vision-Enabled Robotic Platform (MDPI, 2026)

This work integrates a YOLOv8-based perception module with an Intel RealSense depth camera, Raspberry Pi 5, and a SCORBOT-ER 4U robotic manipulator for pick-and-place sorting. Objects were detected with confidence scores exceeding 0.71, and the full pipeline — from detection to robotic actuation — achieved an overall sorting accuracy of approximately 81.8% across multiple test batches. The authors present the system as a scalable proof of concept for controlled-environment automated segregation, while noting that performance in cluttered or unstructured scenes remains a challenge.

2.4 Multi-Class Waste Segregation Using Computer Vision and Robotic Arm (PeerJ Computer Science, 2024)

This study builds a multi-class dry-waste segregation prototype using the YOLOv5 object-detection model connected to a 3D-printed robotic arm. The system targets four dry-waste categories — paper, plastic, metal and glass — and correctly segregated items in 8 out of 10 controlled trials. The authors emphasise that separating dry waste into finer sub-categories (rather than only wet/dry) is important for maximising material-recovery value, though their trial success rate indicates room for improvement in detection robustness.

2.5 An Intelligent Robotic and Bio-Digester Framework for Smart Waste Management (2026)

This paper combines a YOLOv8-based robotic segregation module (using a MyCobot 280 Jetson Nano arm with ROS-based path planning) with a downstream bio-digester for biodegradable waste, optimised using a Particle Swarm Optimisation (PSO) algorithm and regression model. The integrated system classifies waste into four categories with a reported sorting accuracy of 98% and demonstrates efficient biological conversion of the segregated organic fraction, illustrating how segregation output can feed directly into downstream recycling or energy-recovery processes rather than stopping at classification alone.

2.6 A Comprehensive Review of AI-Powered Waste Segregation: Advancing the Vision of Smart Cities (ScienceDirect, 2026)

Rather than proposing a new system, this review synthesises the broader literature on AI, IoT, blockchain and cybersecurity applications in waste segregation. It reports that 80–85% of surveyed studies achieve classification accuracies above 90%, but fewer than 15% address real-world scaling, dataset localisation, or long-term deployment. The review highlights persistent barriers: high cost of robotic deployment, cultural resistance to source-level segregation, and a shortage of studies validating systems outside controlled laboratory settings — findings that directly inform the research-gap analysis in Section 5.

3. Comparative Analysis of Reviewed Studies

Table 1 compares the six reviewed studies on research problem, methodology, dataset/tools, key findings, limitations and research gap, as required by the literature-survey format.

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations / Gap
Sci. Reports, 2025	Multi-class waste image classification	ResNet-based CNN with data augmentation	15,535 images, 12 classes	98.16% accuracy; strong generalisation	Dataset still lab-curated; no robotic actuation
WJEE, 2024	Point-of-disposal	VGG16 CNN +	Wet / dry / e-waste	98% sorting	Only 3 broad

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations / Gap
	waste classification	Raspberry Pi 4 + camera	images	efficiency at bin level	categories; no analytics layer
MDPI, 2026	Real-time robotic sorting	YOLOv8 + DNN + SCORBOT robotic arm	RealSense depth camera, embedded hardware	81.8% end-to-end sorting accuracy	Struggles with cluttered / unstructured scenes
PeerJ CS, 2024	Fine-grained dry-waste segregation	YOLOv5 + 3D-printed robotic arm	Paper, plastic, metal, glass items	8/10 correct trials in controlled tests	Small trial size; moderate detection robustness
arXiv, 2026	Segregation + downstream processing	YOLOv8 + ROS arm + PSO bio-digester	4-category classifier, sensor-fed digester	98% sorting accuracy; efficient bio-conversion	High system complexity and integration cost
ScienceDirect review, 2026	Synthesis of AI/IoT waste segregation literature	Systematic literature review	Cross-study meta-analysis	80–85% of studies exceed 90% accuracy	<15% address real-world scaling or localisation

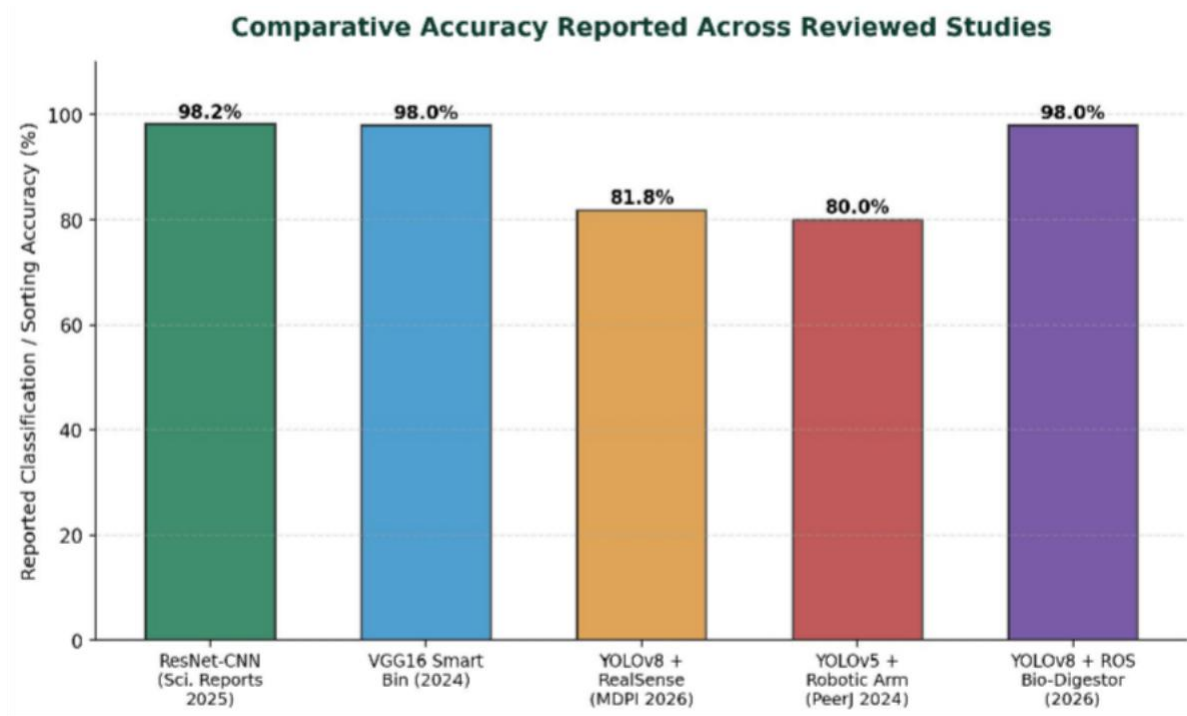


Figure 2. Reported classification / sorting accuracy across the reviewed studies.

4. Limitations and Challenges of Existing Approaches

- **Dataset diversity:** Most models are trained on relatively clean, well-lit, lab-curated images and are not validated on cluttered, overlapping, or damaged waste items typical of real collection facilities.
- **Category granularity:** Several systems classify waste into only two or three broad categories (wet/dry/e-waste), which is insufficient for maximising material-recovery value compared to fine-grained categories (paper, plastic type, metal, glass).
- **Robotic deployment cost:** High-precision robotic arms and depth-sensing hardware (e.g., SCORBOT, RealSense) raise capital cost, limiting adoption in low-resource or developing-region facilities.

- **Fragmented system design:** Most reviewed papers optimise a single stage — either classification accuracy or robotic actuation — with few studies integrating classification, sorting, IoT monitoring and reporting into one deployable pipeline.
- **Limited real-world validation:** The comprehensive review (Section 2.6) found that fewer than 15% of studies test at scale outside controlled environments, and cultural/behavioural barriers to source segregation are rarely addressed by purely technical solutions.
- **Absence of analytics and reporting:** Few reviewed systems generate recycling-rate statistics or management reports automatically; most stop at the point of physical sorting.

5. Research Gap Analysis

Synthesising the six reviewed studies against the assigned problem statement reveals a consistent gap: no single reviewed system combines all four required capabilities — computer-vision identification, robotic sorting automation, recycling-statistics monitoring, and automated report generation — within one cost-effective, real-world-validated pipeline. High-accuracy classifiers (Sections 2.1, 2.2, 2.5) are rarely paired with robust robotic actuation tested outside controlled settings, while systems with working robotic arms (Sections 2.3, 2.4) report comparatively lower end-to-end accuracy, indicating an unresolved trade-off between classification precision and physical sorting reliability. Furthermore, the analytics/reporting layer that facility managers need for monitoring recycling efficiency is largely absent from the technical literature, even though it is explicitly required by the assigned problem statement.

6. Proposed System Overview

Building on the strengths and gaps identified above, the proposed system layers a lightweight CNN/YOLO classification model (informed by the high accuracy of Sections 2.1 and 2.2) on an edge device, paired with a moderately priced robotic actuation stage (informed by the practical deployments of Sections 2.3 and 2.4), and adds the analytics and reporting layer identified as missing in Section 5. Figure 3 shows the four-layer architecture, and Figure 4 shows the corresponding operational pipeline.

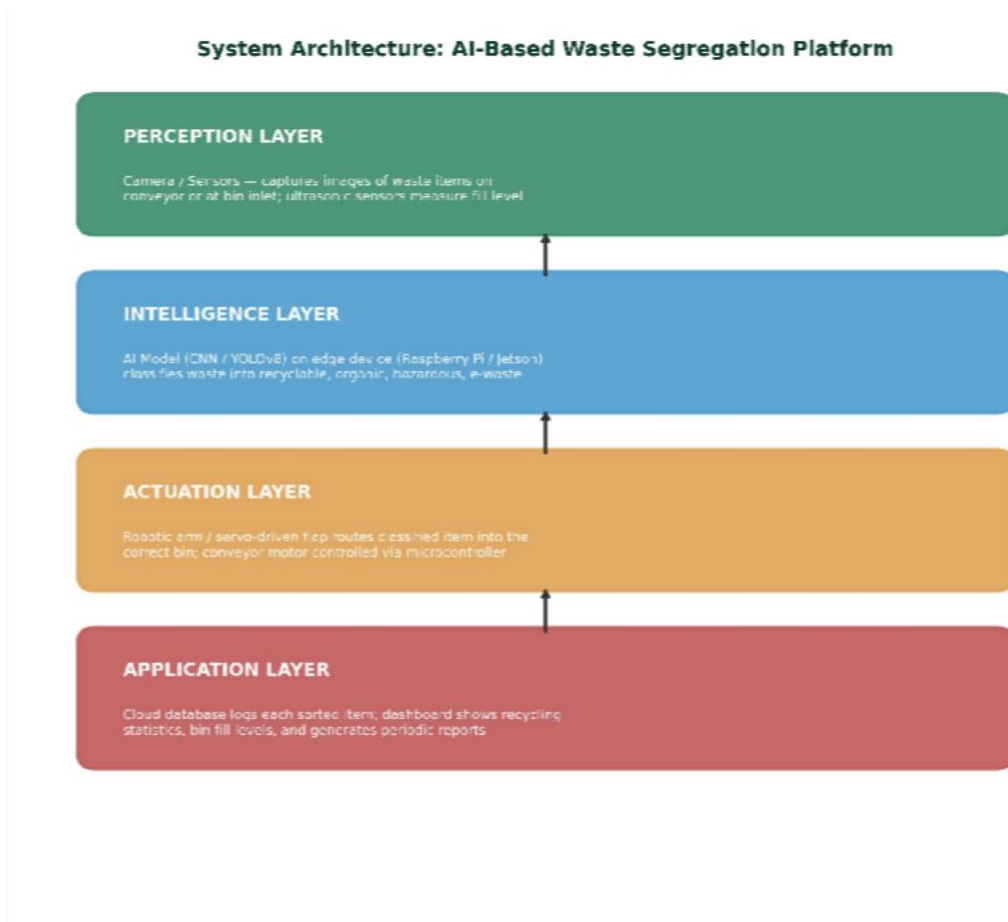


Figure 3. Proposed four-layer system architecture: perception, intelligence, actuation, and application layers.

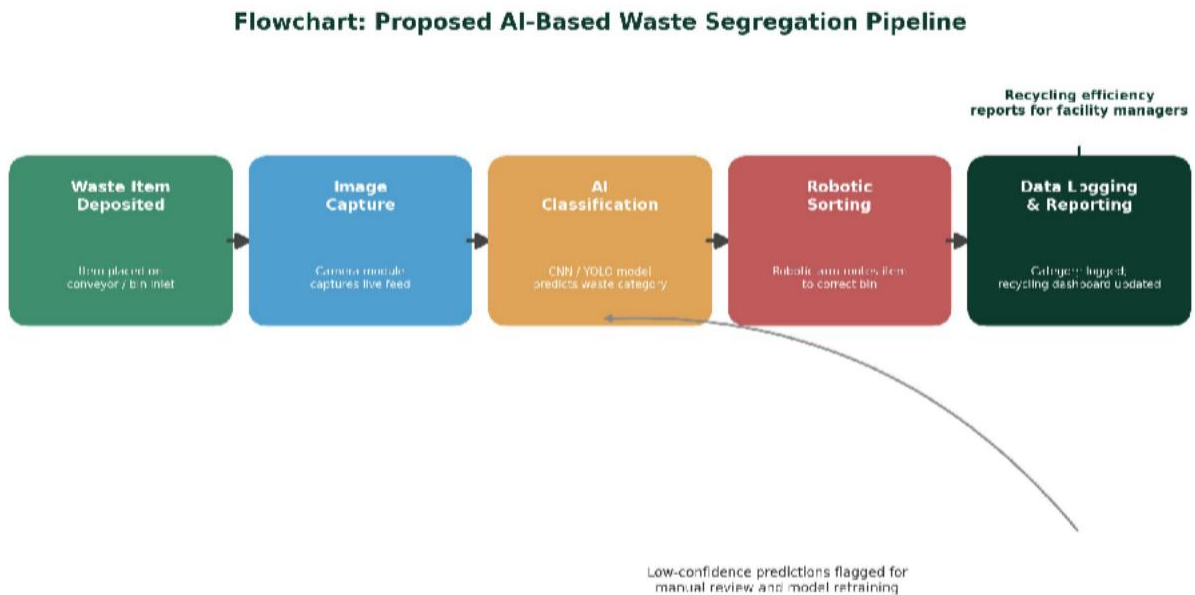


Figure 4. Flowchart of the proposed AI-based waste segregation and reporting pipeline.

7. Future Research Directions

- Build and publicly release cluttered-scene, real-facility waste datasets (extending efforts such as the ZeroWaste dataset) to close the lab-to-field generalisation gap.
- Investigate lightweight, low-cost robotic actuation (servo-driven flap/chute mechanisms) as a middle ground between simple smart bins and industrial robotic arms, to reduce deployment cost for developing regions.
- Develop end-to-end pipelines that combine classification, sorting, and a reporting dashboard as a single evaluated system, rather than evaluating each stage in isolation.
- Incorporate active-learning or human-in-the-loop review for low-confidence predictions, allowing continuous model retraining from real deployment data.
- Study behavioural and policy interventions alongside technical deployment, since the reviewed literature indicates that source-level segregation habits materially affect downstream AI performance.
- Explore integration with downstream processing (e.g., bio-digestion, as in Section 2.5) so that segregation output directly improves measurable recycling or energy-recovery outcomes.

8. Conclusion

The reviewed literature confirms that AI-based computer vision and robotic sorting can substantially improve waste-segregation accuracy and consistency over manual methods, with several studies reporting accuracies above 95% under controlled conditions. However, no reviewed system fully satisfies the assigned problem statement's requirement for an integrated pipeline spanning identification, automated sorting, recycling-statistics monitoring, and report generation, and real-world validation beyond laboratory settings remains limited. The proposed four-layer architecture directly targets this integration gap, and the future directions outlined above provide a concrete roadmap for developing a deployable, cost-effective, and responsibly evaluated AI waste-segregation system.

References

- [1] Intelligent waste sorting for urban sustainability using deep learning, Scientific Reports, 2025.
- [2] AI-Based Smart Bin for Efficient and Sustainable Waste Classification, World Journal of Environmental Engineering, 2024.
- [3] Real-Time Solid Waste Sorting Using a Vision-Enabled Robotic Platform, MDPI Robotics, 2026.
- [4] Lahoti, J., Sn, J., Krishna, M.V., Prasad, M., BS, R., Mysore, N., Nayak, J.S. Multi-class waste segregation using computer vision and robotic arm, PeerJ Computer Science, 2024.
- [5] Khatri, R., Tewari, A., Sharma, N., Srinivas, M.B. An Intelligent Robotic and Bio-Digester Framework for Smart Waste Management, arXiv, 2026.
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